

Appendix A: Cutting a Helix Auger

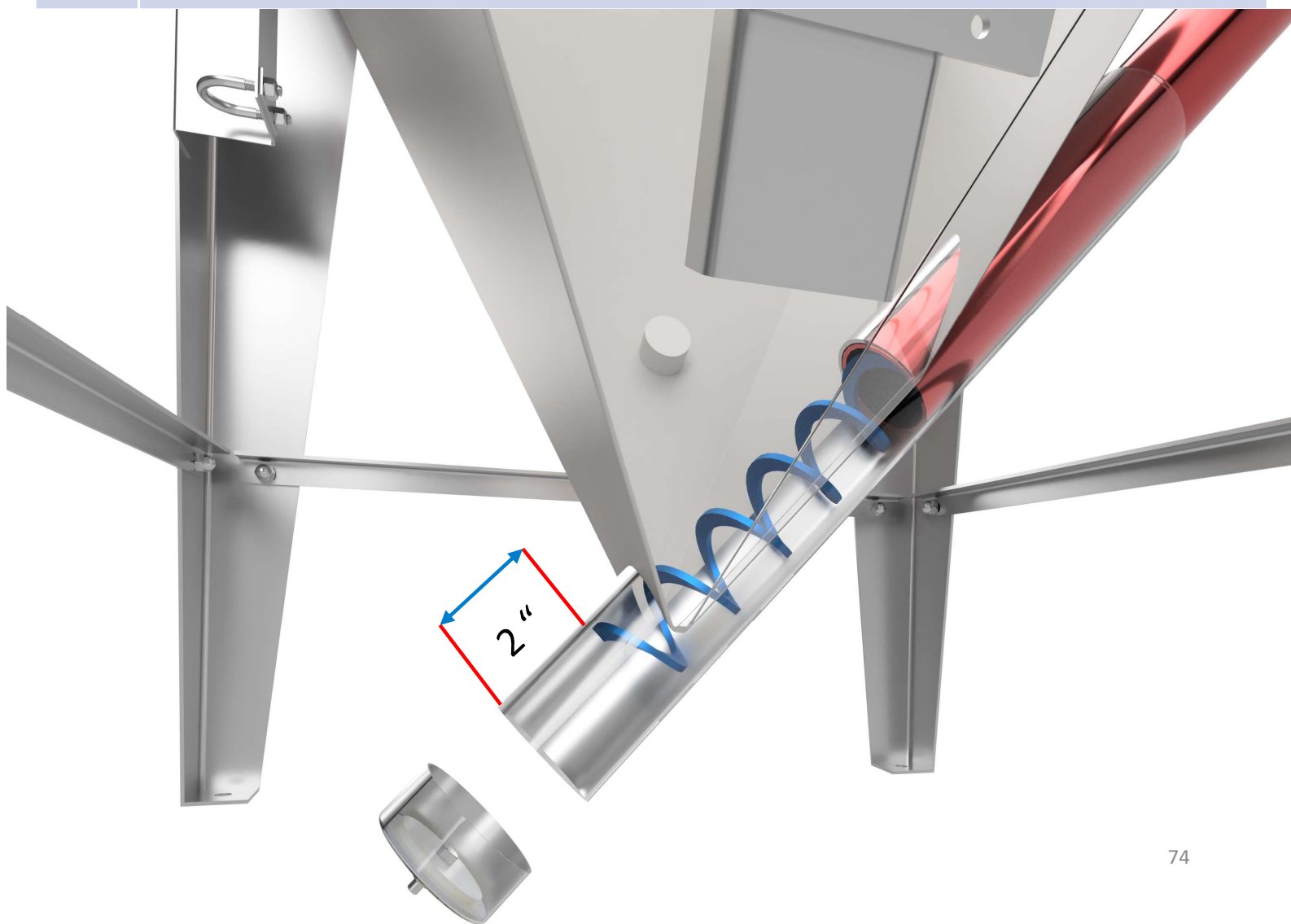


Prevent serious injury or death.

**Wear personal protective devices (PPD) appropriate for the job.
Move object being cut to a designated safe work location for cutting/grinding.**

The trimming or cutting of a Helix™ auger is a common maintenance practice as the augers can stretch over time. Also, at installation if distances between the hopper and the discharge need to be modified it may require shortening the auger. Below are the steps to properly trim a 250-300 series auger.

1. The desired operational distance between the Helix auger and the end cap on the hopper boot is 2 inches.



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Prevent serious injury or death.

**Wear personal protective devices (PPD) appropriate for the job.
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2.

With the auger inserted into the casing push it up so that it is contacting the stepped part of the connector in the drive box.



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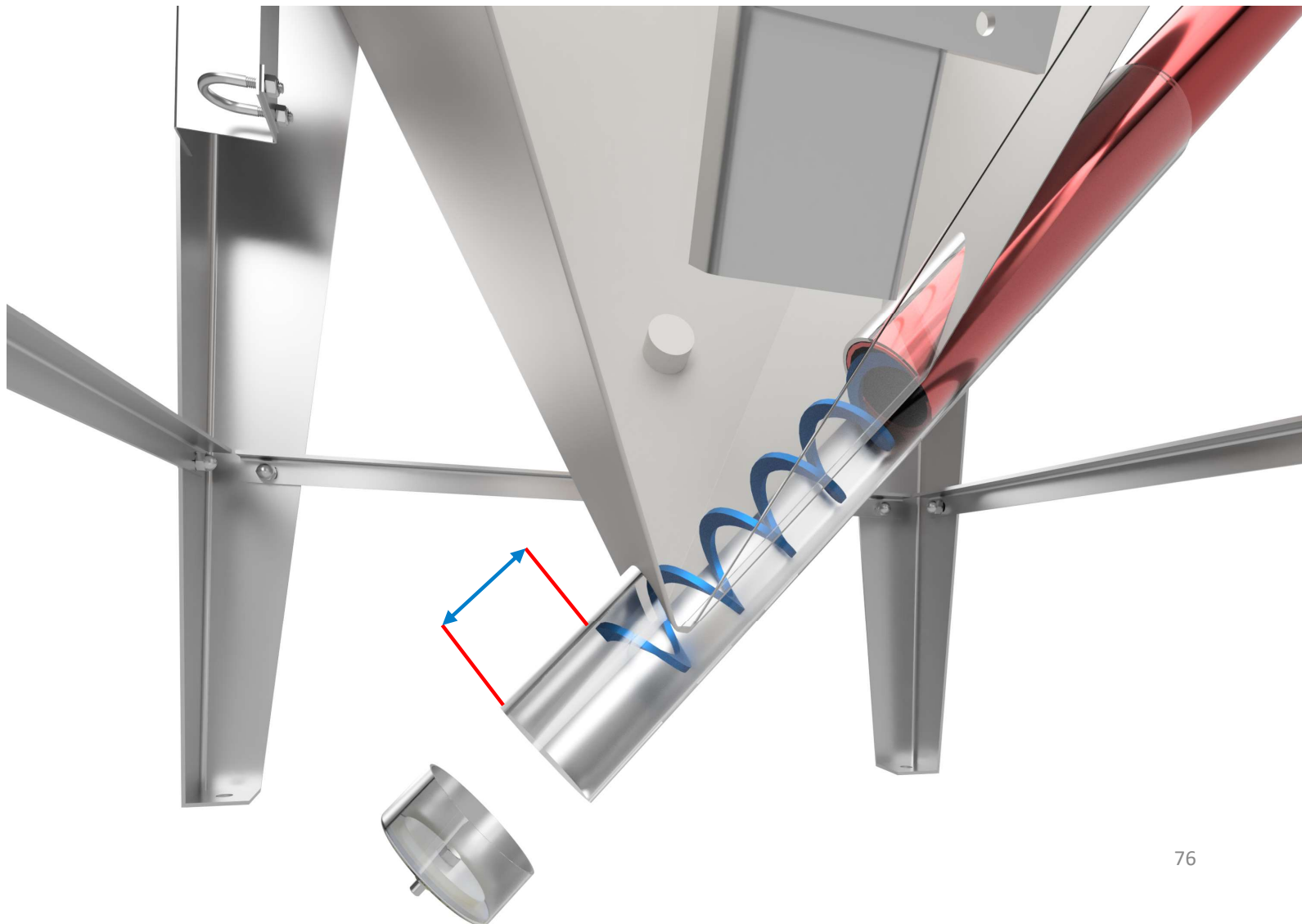


Prevent serious injury or death.

**Wear personal protective devices (PPD) appropriate for the job.
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3.

Now measure the distance from the bottom of the auger to the end of the hopper boot.



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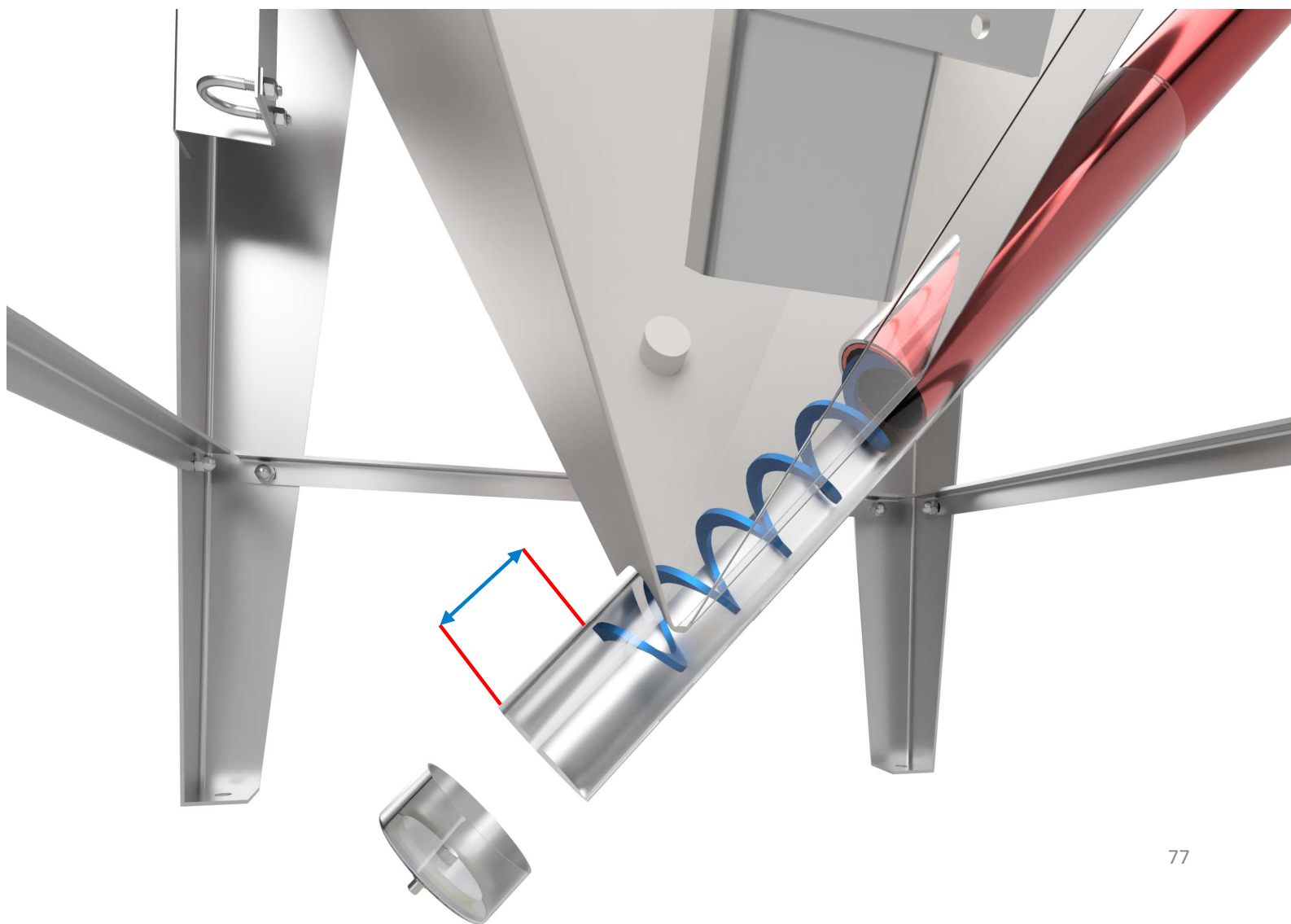
Prevent serious injury or death.

**Wear personal protective devices (PPD) appropriate for the job.
Move object being cut to a designated safe work location for cutting/grinding.**

4. Take your measured dimension and subtract it from the 2-inch standard. The difference will be how much needs to be removed from the auger.

NOTICE

Example: If you measured and the auger was a 1/2 inch from the end cap you would subtract that from 2 inches. The difference of 1-1/2 inches would be what needs to be trimmed from the auger.



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5. Once you have your measurement, remove the auger from the casing
6. Cut at the top (drive side) of the auger.

NOTICE

Cutting the top (drive side) of the auger is only necessary on augers that have a closed end on the hopper side of the auger. Augers that are open on both ends can be trimmed at the top or bottom.

NOTICE

Make sure the cut is square and all burrs and sharp edges have been removed from the auger before reinstalling.

Open End of Auger (Drive Side)



Closed End of Auger (Hopper Side) Do Not Cut

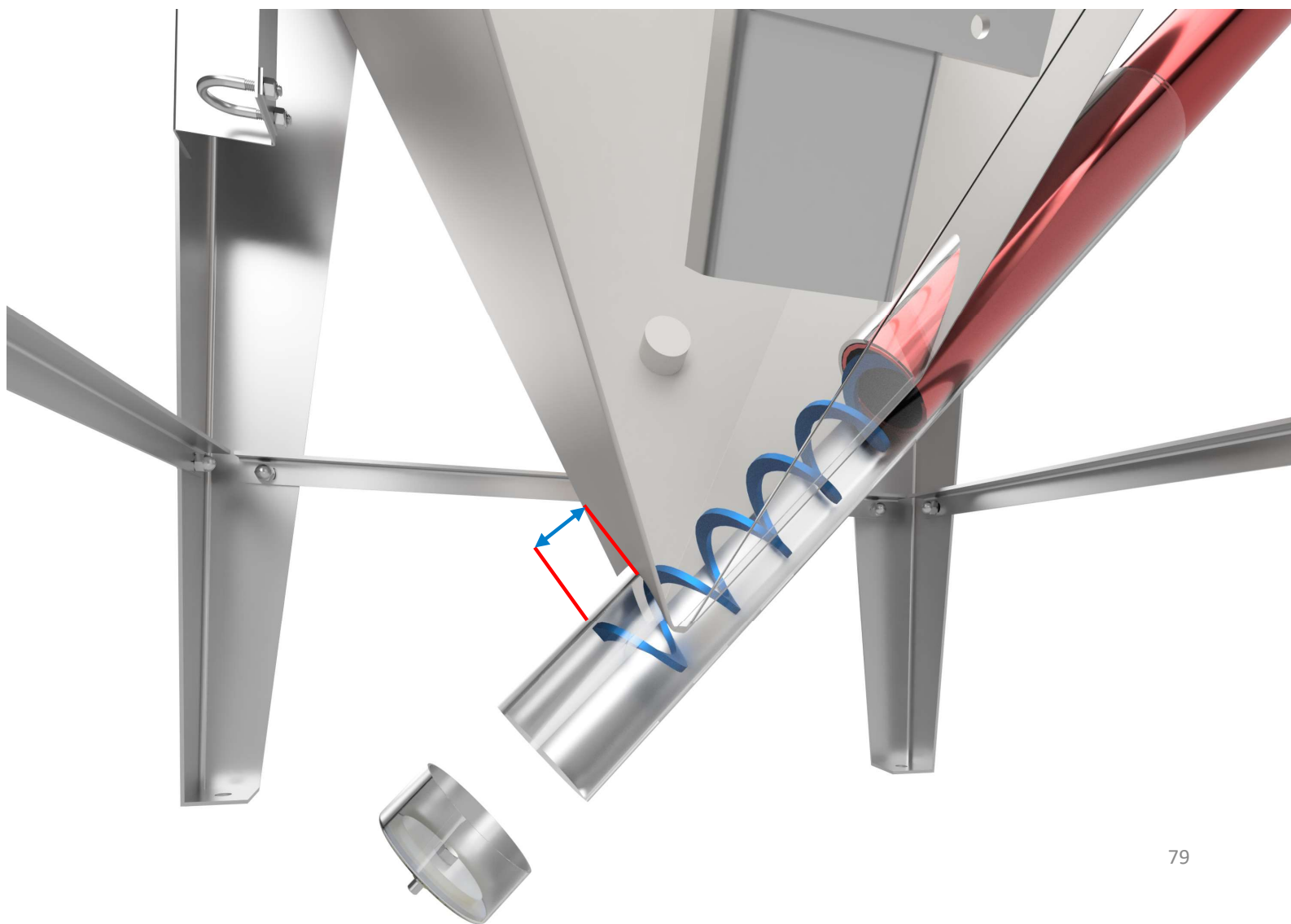


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7. Reinstall the auger and check length as called out on pages 74-76
8. If more trimming is needed repeat the steps from the previous pages until desired length is reached.

NOTICE

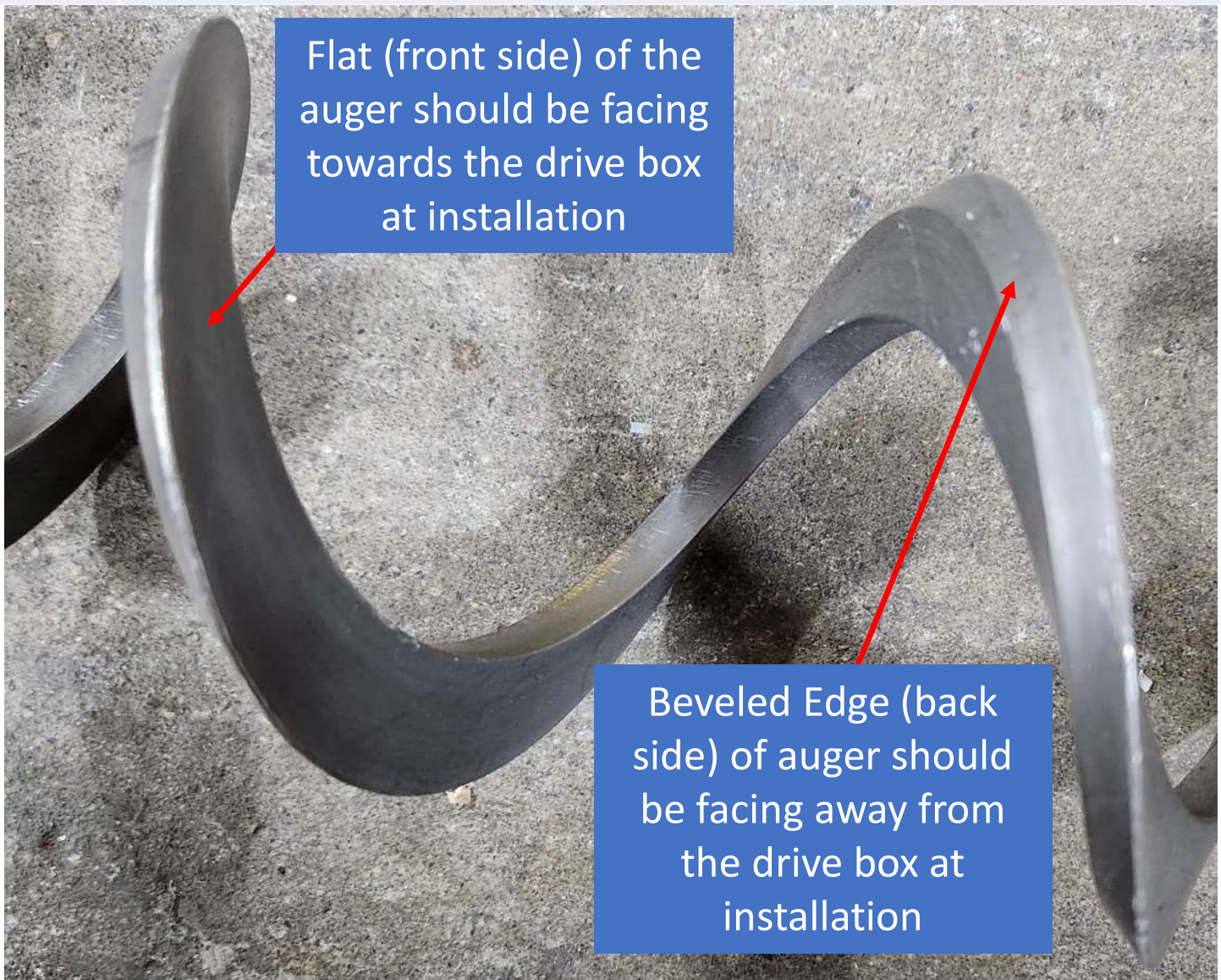
At minimum there must be at least one full auger flight length past the hopper wall present. Any shorter than that and the auger could come out of the boot while running. This could lead to unwarranted damage of the auger or hopper.



Appendix B: Beveled Edge Auger Installation

The beveled edge auger has a mitered cut on the trailing edge of the screw. The reduced surface area of the outer edge along with the beveled trailing edge reduces material build up within the casing. Because of this feature the auger must be installed in the correct orientation to operate properly.

1. Identify the flat (front side) of the auger and the beveled (back side) of the auger
2. Orient the auger so the flat side (front side) of the auger is facing towards the drive box at installation.
3. Install the auger as detailed on pages 38-44



Appendix B: Beveled Edge Auger Installation

Below are two images showing the beveled edge auger from two different installation positions. For 250-300 series Helix conveyors most installations will be from the end cap. Some installations could be from the drive if space at the rear of the conveyor is limited.

1.	View from the back side of the auger looking towards the drive. This is how the auger should look when being installed through the end cap up towards the drive.
2.	View from the front side of the auger looking down towards the hopper. This is how the auger should look if being installed from the drive box down to the hopper.



1. This is how the auger should look when being installed through the end cap towards the drive.



2. This is how the auger would look if being installed from the drive towards the hopper.

Appendix C: Helix Hopper Agitator

Agitator hoppers are utilized on Helix conveyors when dealing with materials that bridge, rat hole, or are sluggish. There are four styles of agitators offered by Hapman.

Flat Wire: This agitator serves two primary functions. The first is to keep the material agitated to prevent bridging in the hopper. The second is to drive sluggish material towards the Helix™ auger inlet to ensure there is a steady rate of material.

Round Wire: This style of agitator is used with heavier bulk density materials to mitigate bridging in the hopper. This design reduces the amount of surface drag compared to the flat wire model. Allowing it to move easily through the heavier materials, to keep them flowing. This agitator will still move material towards the Helix™ auger inlet but at a reduced rate compared to the flat wire style.



Flat Wire Agitator



Round Wire Agitator

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Pin: The pin style agitator is used in applications where the material when agitated, will flow on its own.

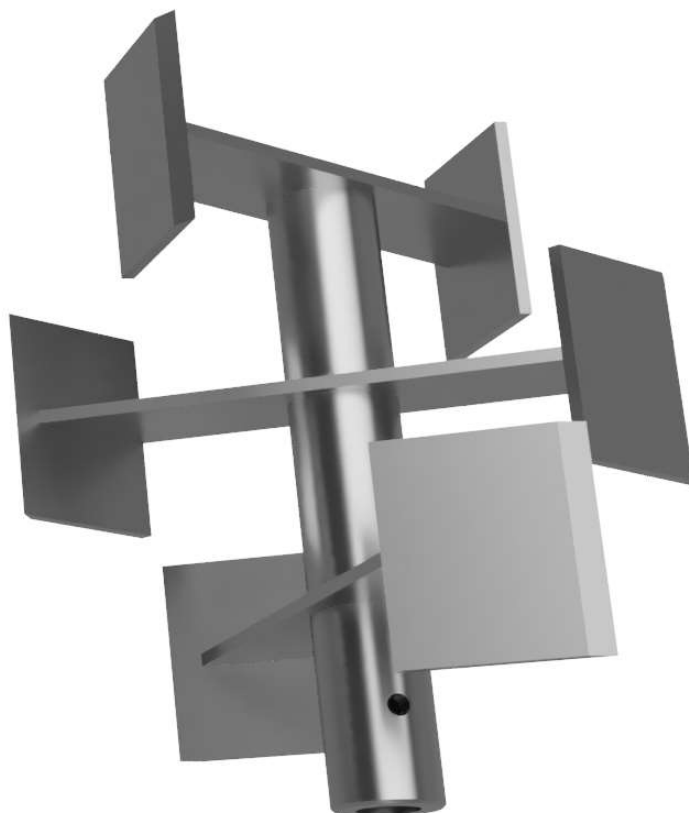
Paddle: This agitator is designed to cut through materials that like to conglomerate once introduced into the hopper.

NOTICE

The pin and paddle style agitators can be retrofitted to most existing hoppers if needed.



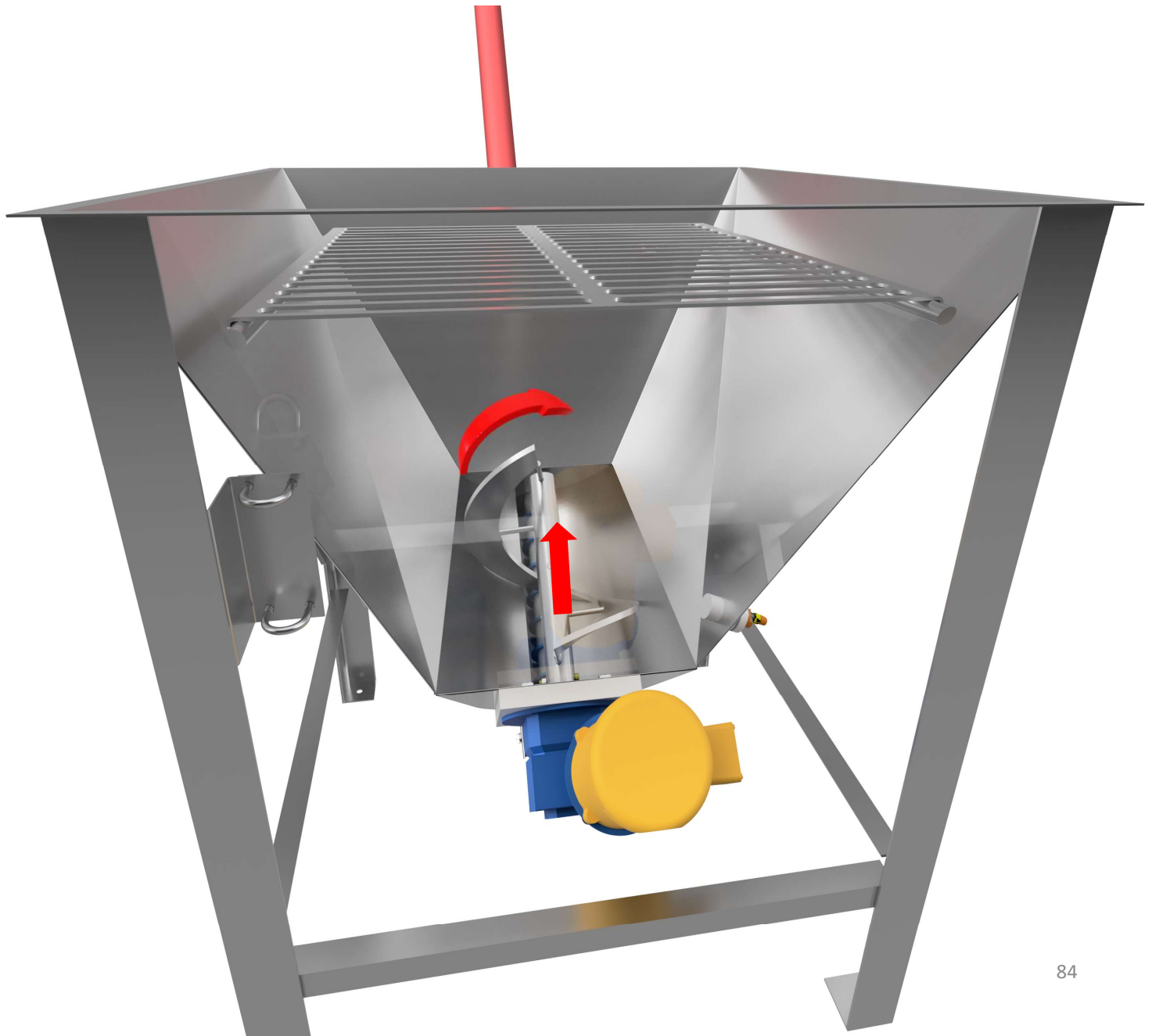
Pin Agitator



Paddle Agitator

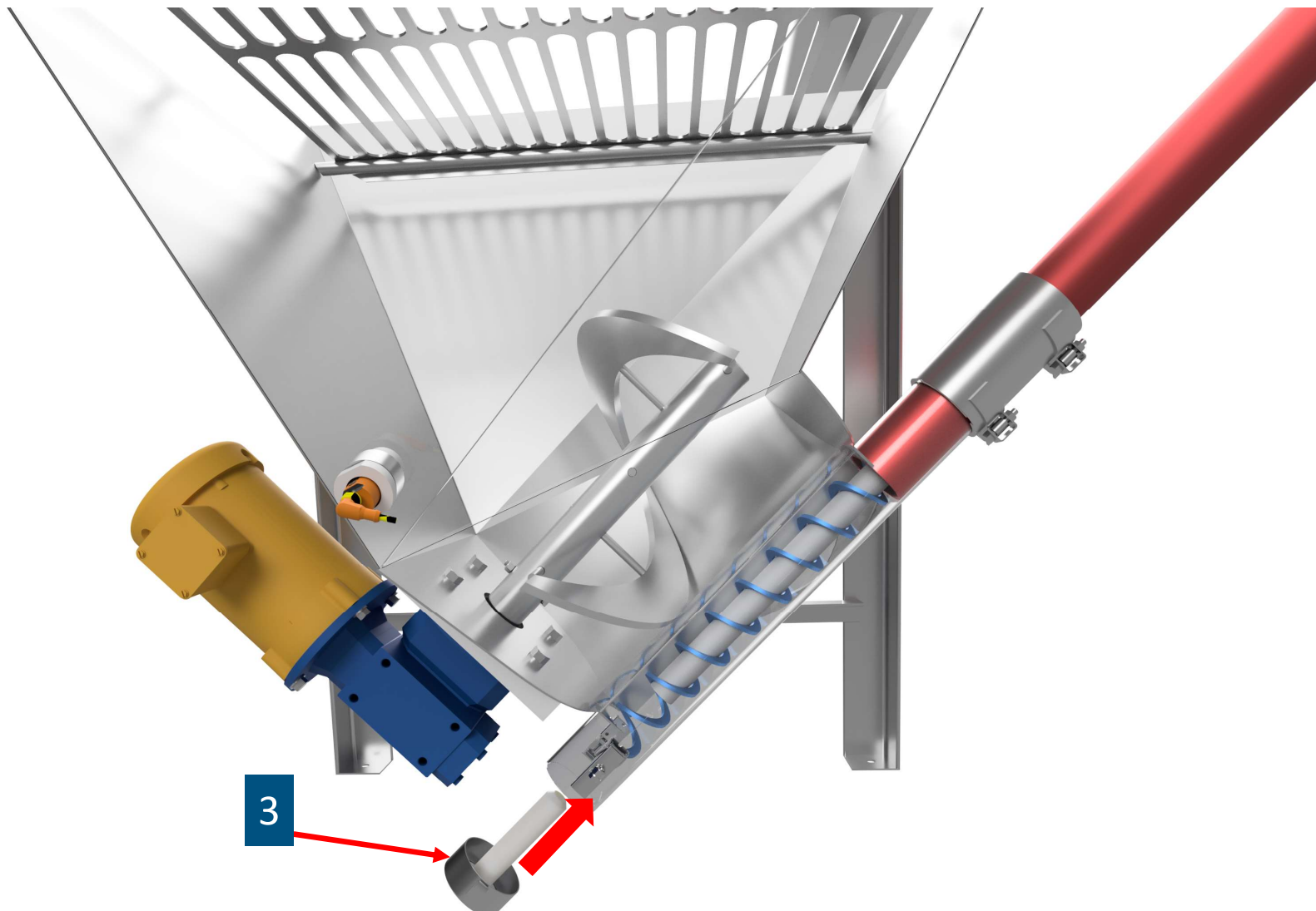
Appendix C: Helix Hopper Agitator

1. The agitator should rotate counterclockwise. (*This refers to the agitator inside the hopper, not the motor*) When running, the ribbon and round wire agitators should appear to be moving towards the Helix™ inlet.
2. Ribbon and round wire agitators should cycle intermittently during operation to avoid over filling of the Helix™ auger inlet. Pin and paddle style agitators can run continuously.



Appendix C: Helix Hopper Agitator

3.	On agitator hoppers Hapman includes a UHMW probe that attaches to the end cap. This probe helps keep the rear of the Helix™ auger centered and helps prevent the auger from rising and becoming entangled with the agitator.
4.	The probe should slide smoothly up through the inside of the auger
5.	Then the end cap will be latched in place.



Appendix C-2: Helix Hopper Agitator

Below are the steps for removal of the hopper agitator assembly and drive.



Severing and entanglement hazard.

Conveyor may start automatically or from a remote location.

Keep clear of conveyor and all moving parts.

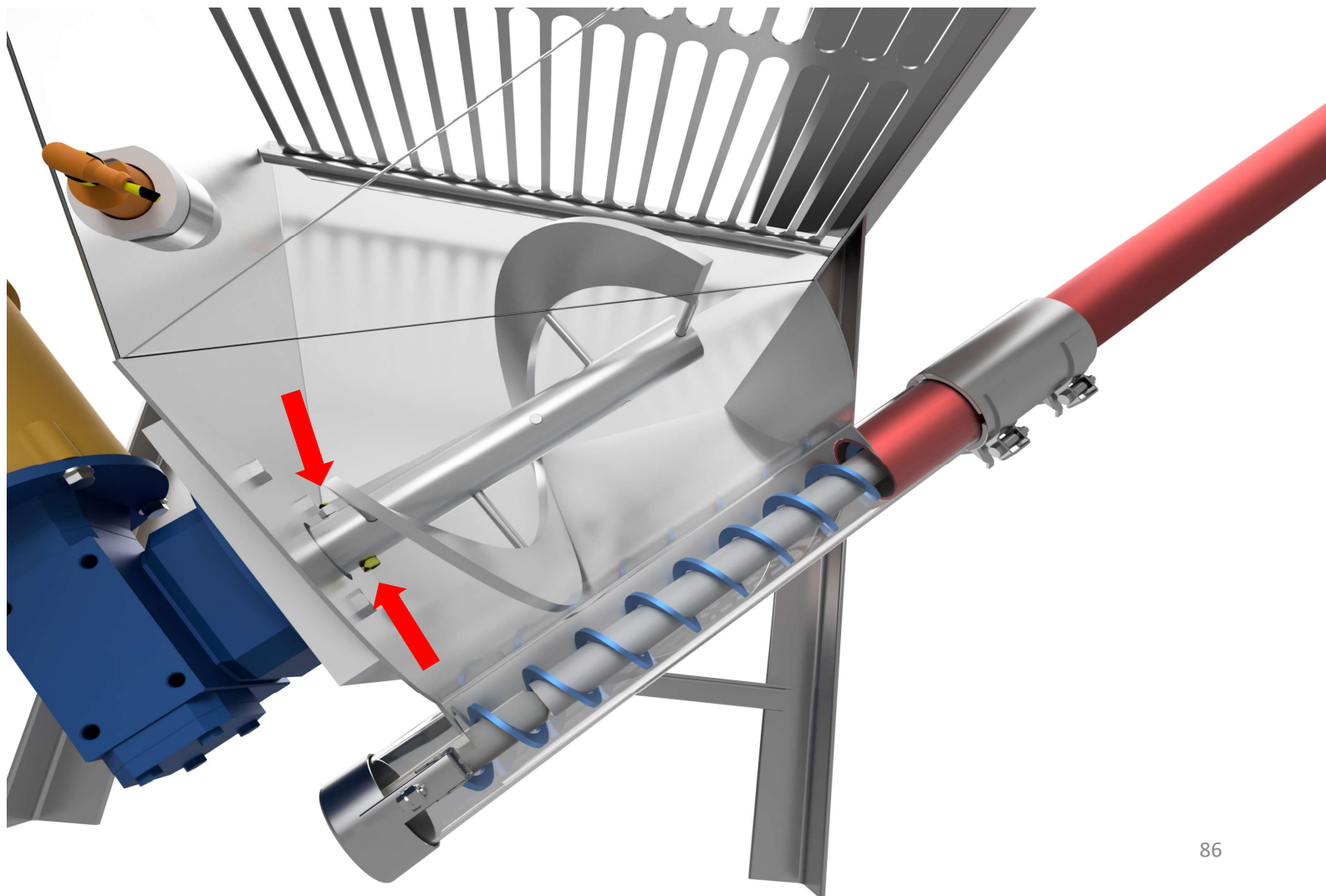
Disconnect and lockout power source before adjusting, cleaning, unclogging or servicing.

1.

Lock out-Tag out control panel and any local disconnects, so agitator cannot be turned on during disassembly/assembly.

2.

Remove all material from the hopper and wipe down the agitator so shoulder bolt and nut can be clearly seen.

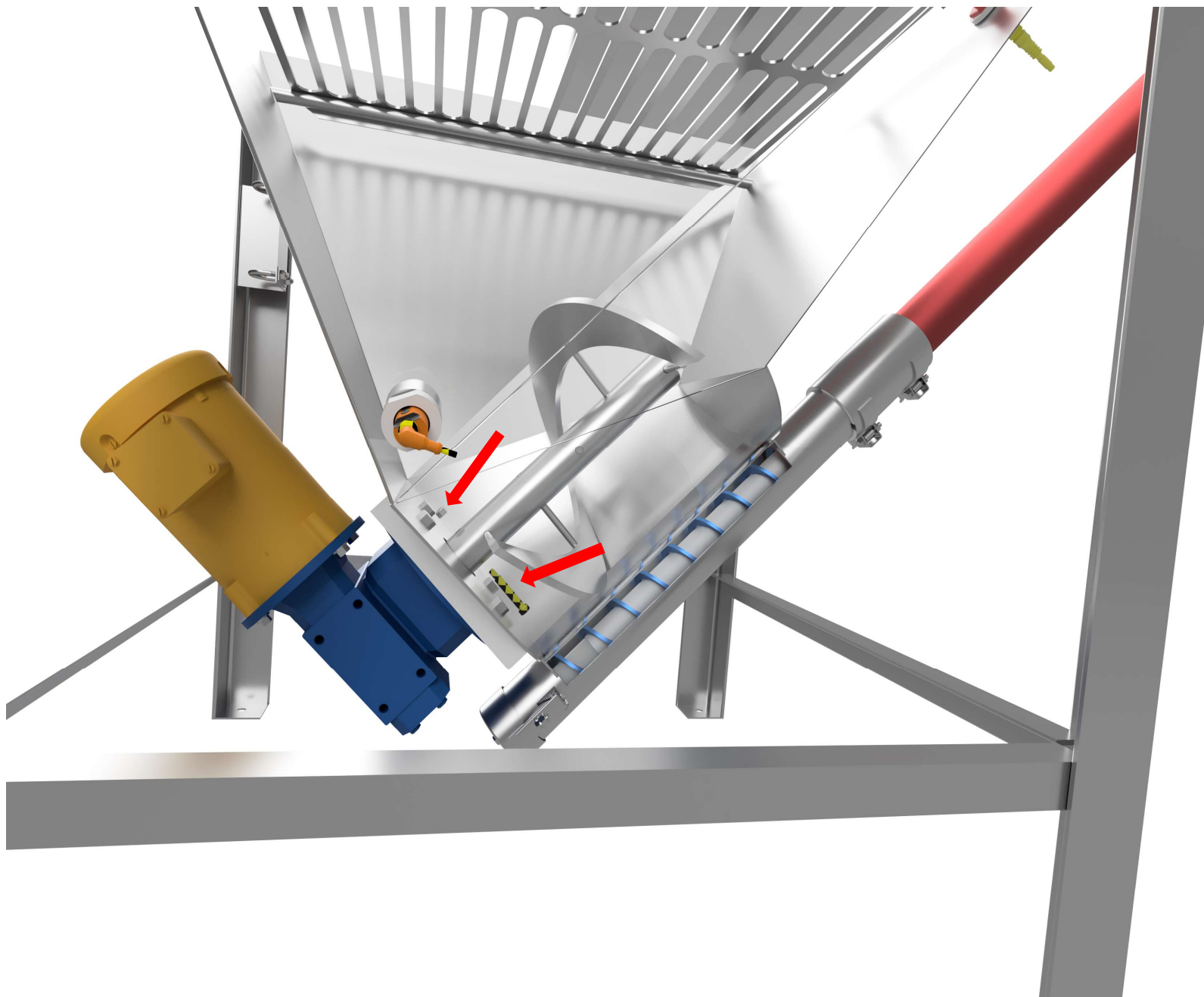


Appendix C-2: Helix Hopper Agitator

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| 3. | Remove nut holding shoulder bolt in place |
| 4. | Remove shoulder bolt |

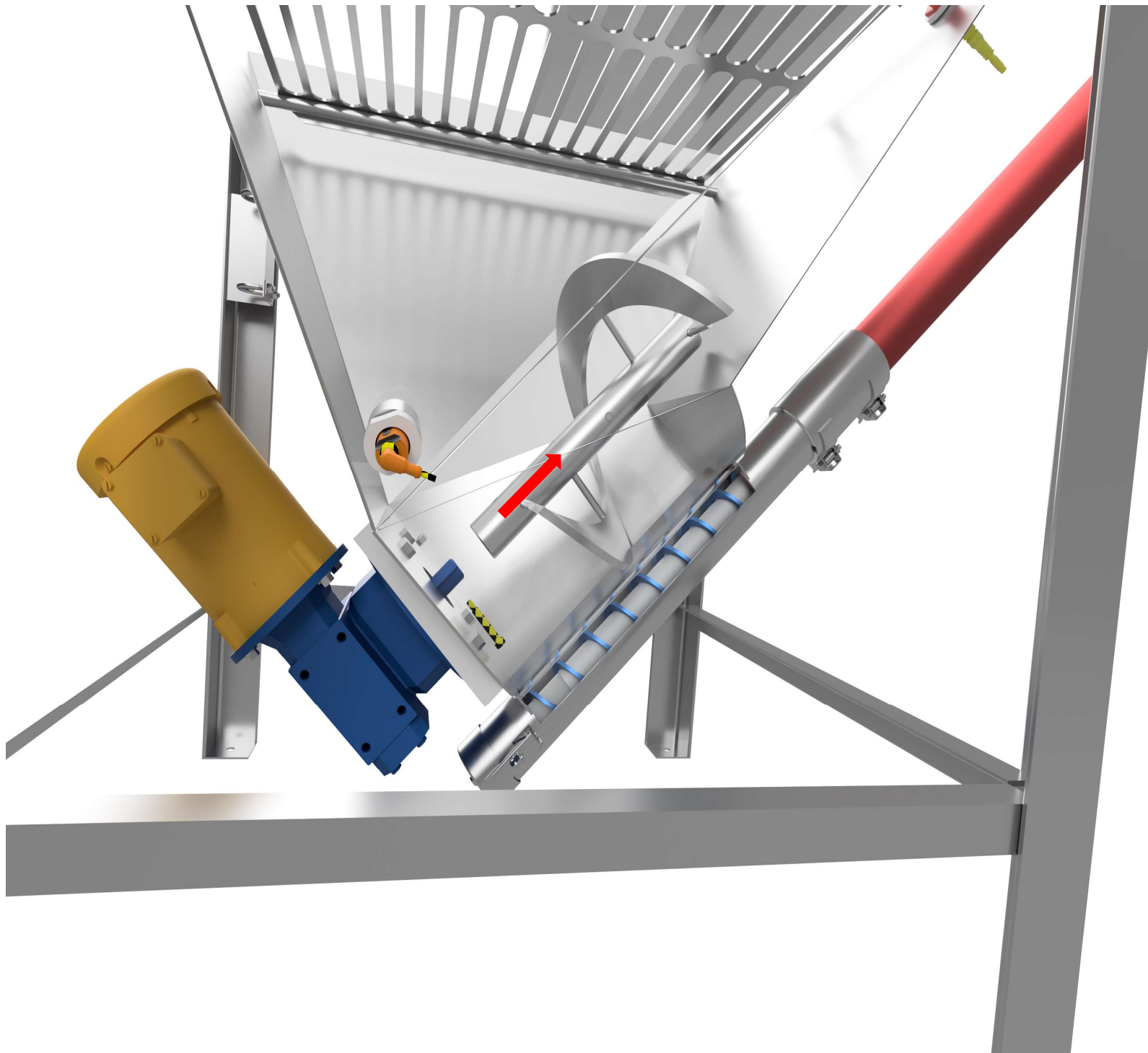
NOTICE

Shoulder bolts that have been in place for a long time may require a punch tool to push them out. For this reason, it is recommended to keep a spare shoulder bolt and nut on hand.



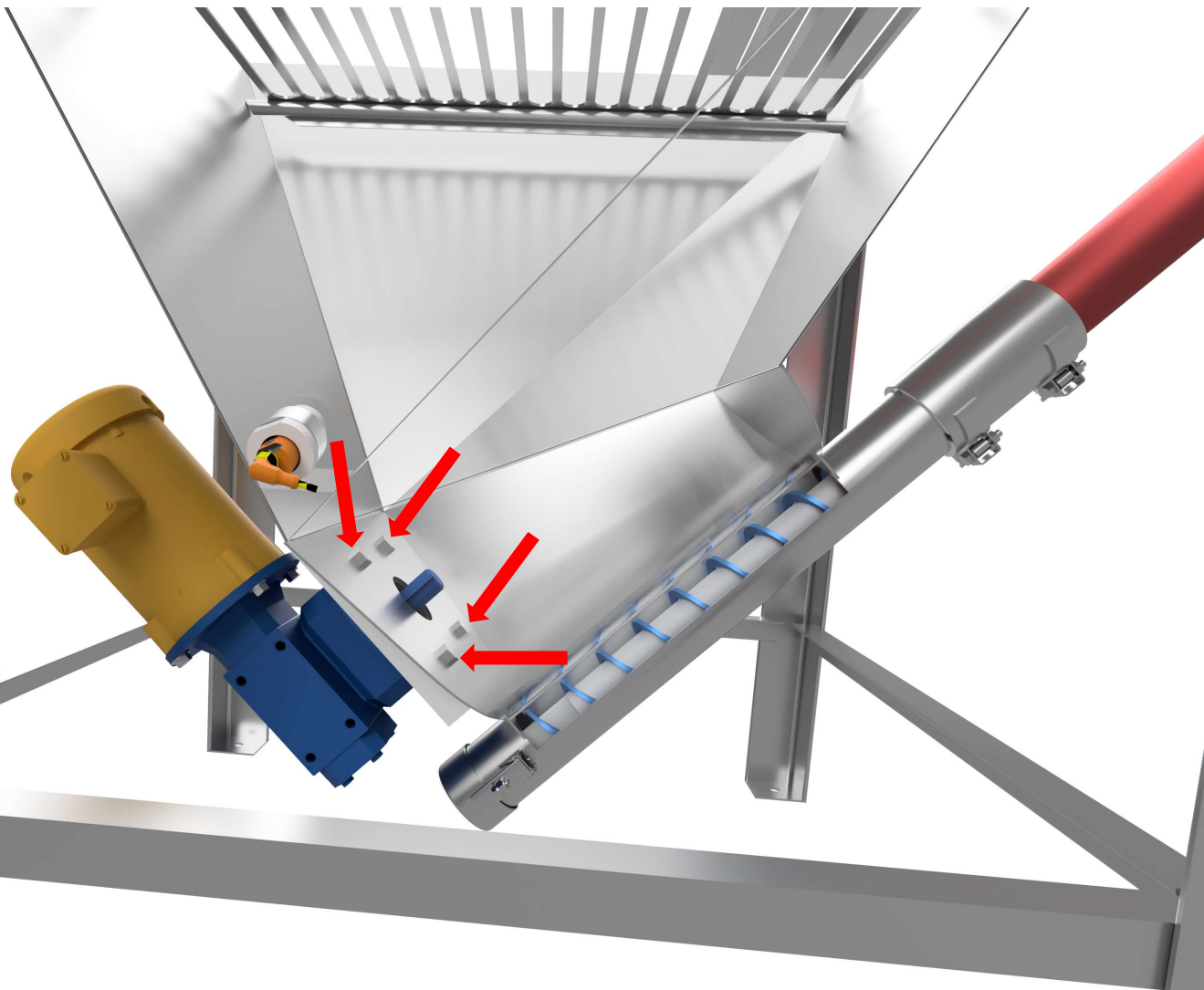
Appendix C-2: Helix Hopper Agitator

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| 5. | Slide agitator assembly towards the Helix drive. |
| 6. | Remove agitator auger assembly from hopper. |



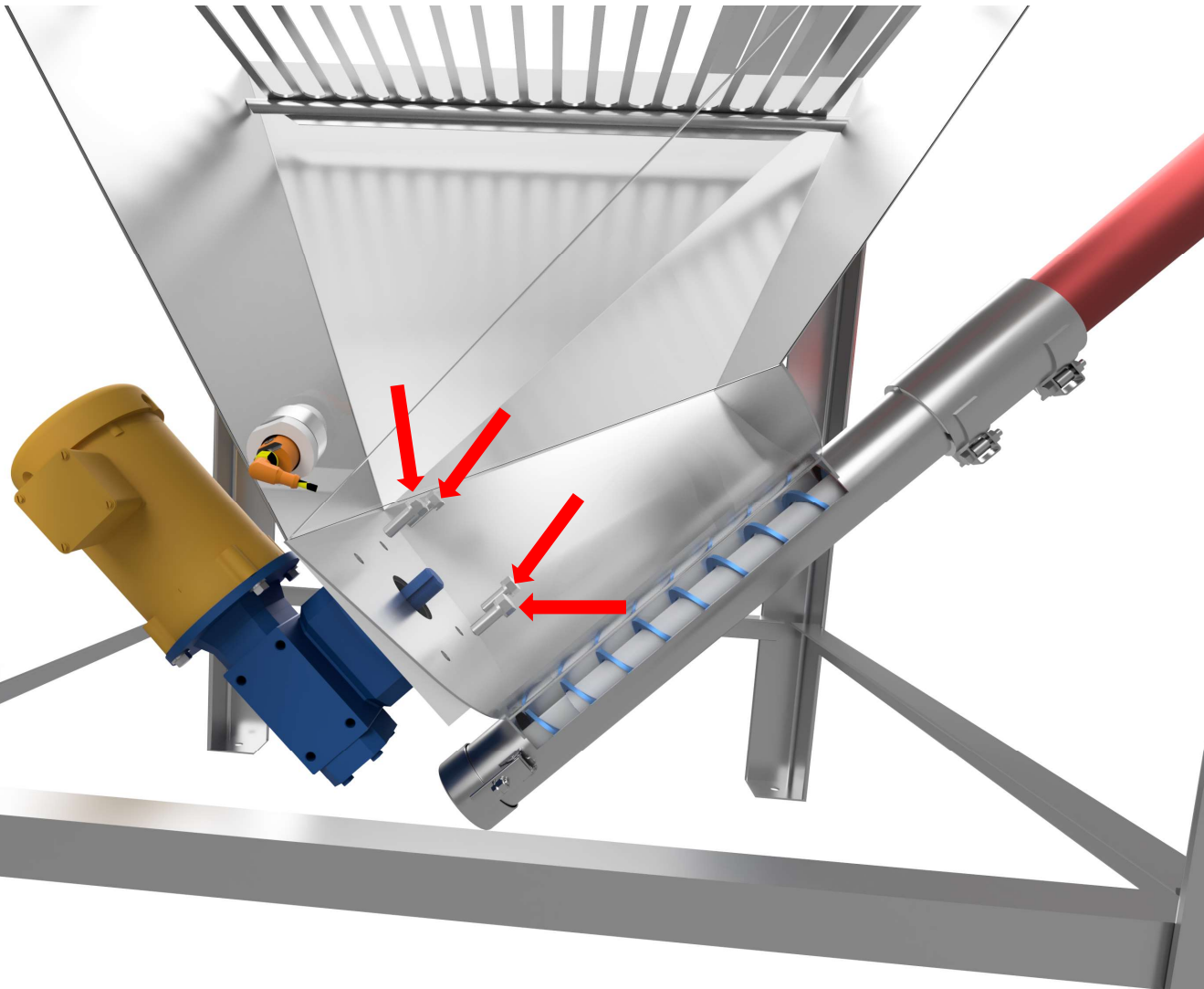
Appendix C-2: Helix Hopper Agitator

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|----|---|
| 7. | Inspect the mounting plate bolt heads for any corrosion or damage |
| 8. | The agitator drive will need to be supported for the next step. Make sure you have something in place that can support and hold the load of the gearbox, motor, and mounting plate. |



Appendix C-2: Helix Hopper Agitator

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| 9. | Remove the mounting plate bolts |
| NOTICE | Make sure you have something in place that can support and hold the load of the gearbox, motor, and mounting plate. |

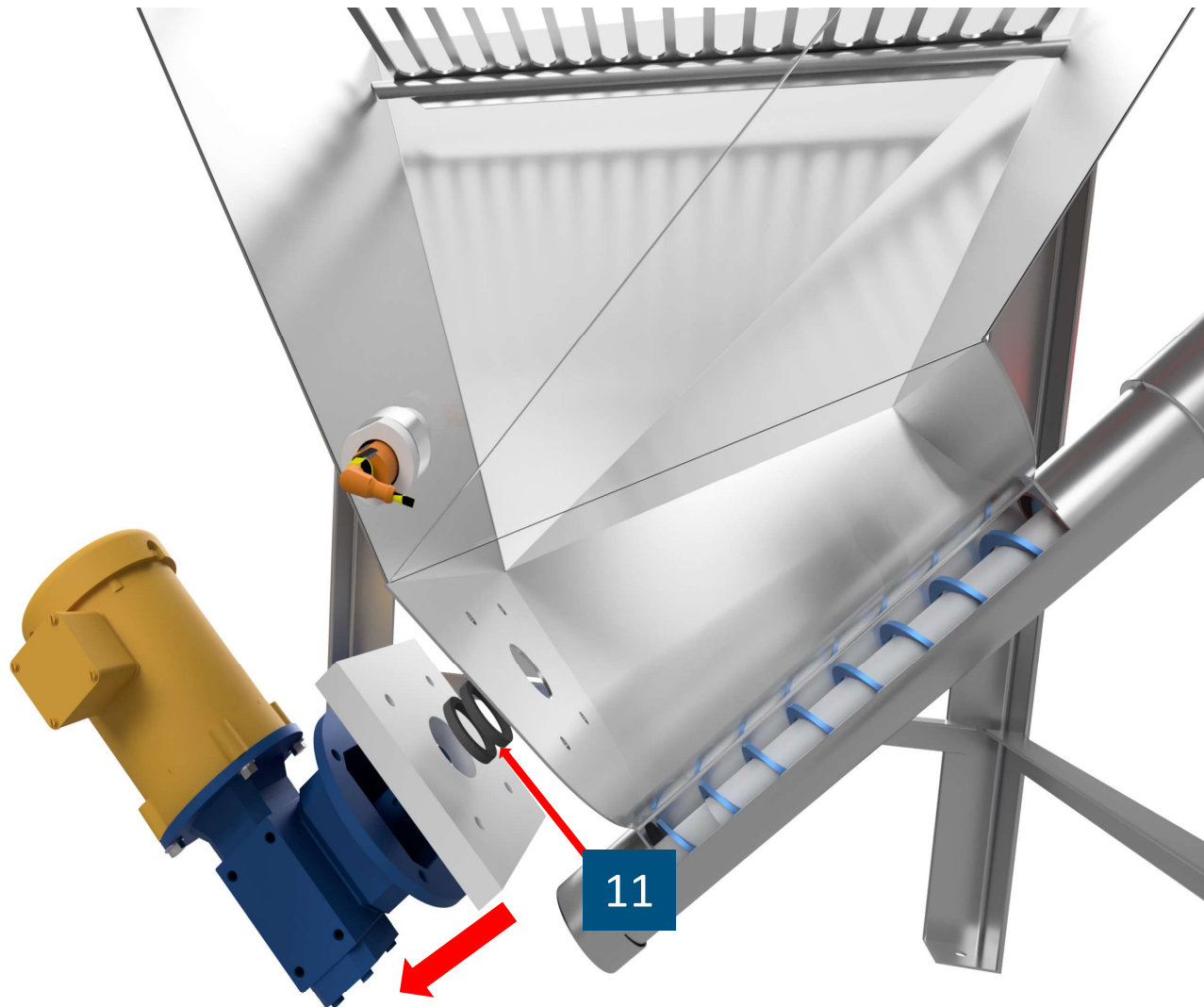


Appendix C-2: Helix Hopper Agitator

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| 10. | Pull mounting plate and drive assembly out of the hopper |
| 11. | Inspect seals for any wear or damage |

NOTICE

It is recommended to replace both lip seals anytime the drive assembly is fully removed from the hopper.



C-2: Recommended Mechanical Spare Parts (Hopper Agitator)

<u>Qty</u>	<u>Description</u>
1	Agitator Auger Assembly
1	Shoulder Bolt w/ nut
1	Motor
2	Lip seals
1	Gear Reducer

NOTICE

Non-factory approved parts can cause improper operation and could cause unsafe operating conditions.

Reference your mechanical drawings or contact Hapman Customer Service with any questions.